

Design Team
Brecksville-Broadview Heights High School
2024-2025

MODEL CURRICULUM: LEARNING PROGRESSIONS

August

Topic/ Focus	General Pacing (days - weeks)	State Learning Standards (highlight power standards)	Key Vocabulary	Learning Targets (student friendly) I am learning:	Link to Assessments & Resources
Technology	3-4 Days	1.1. Empowered Learner Students leverage technology to take an active role in choosing, achieving and demonstrating competency in their learning goals, informed by the learning sciences 1.1.d. understand the fundamental concepts of technology operations, demonstrate the ability to choose, use and troubleshoot current technologies and are able to transfer their knowledge to explore emerging technologies.	Technology Communication Transportation Manufacturing Agriculture Construction Energy & Power Biomedical	History of technology Identify and Explain the 7 systems of technology Real World Technology Connections Future Technology Implications/Outcomes	What is Technology Lecture and Discussion Technology Slides
Design Process	3-4 Days	1.4 Innovative DesignerStudents use a variety of technologies within a design process to identify and solve problems by creating new, useful or imaginative solutions. 1.4.a. know and use a deliberate design process for generating ideas, testing theories, creating innovative artifacts or solving authentic problems	Design Process Identify Problem Brainstorm Design Build Test Evaluate Redesign Share Solutions Sketch Research	Explain the steps in the Design Process Use the steps in the Design Process to solve a real world problem problem	Design Process Lecture and Discussion Processes for each step activities
Team Work, Leadership...	2-3 Days	1.7.c Project Teams Students contribute constructively to project teams, assuming various roles and	Team Work Leadership Collaboration Group Work	Explain the importance of teamwork, groups, collaboration, compromise, and difference in skill sets. How to work together to achieve a common goal	Team Building Activity Analysis of Team building activity

		responsibilities to work effectively toward a common goal. 1.7.d Local and Global Issues Students explore local and global issues, and use collaborative technologies to work with others to investigate solutions.	Compromise		Teamwork and Leadership Lecture, discussion & Videos. Current Skill Set & Interest Student Survey
Problem Solving Real World Connections	2-3 Days	1.3.d Explore Real-World Issues Students build knowledge by actively exploring real-world issues and problems, developing ideas and theories, and pursuing answers and solutions. 1.4.a Design Process Students know and use a deliberate design process for generating ideas, testing theories, creating innovative artifacts or solving authentic problems.	Aptitude Skill Set Passion Desire Motivation Career College	Relate course goal to real world connections such as future careers and college majors	Real World Connections Lecture and Discussion Career, Business, and College Major Research Activity Potential Guest Speakers in Related Areas
Student Driven Projects (# and timing vary based on size and complexity of projects)					
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Identify Problem Stage	Vary Depending on complexity of project	1.5.a Problem Definitions Students formulate problem definitions suited for technology-assisted methods such as data analysis, abstract models and algorithmic thinking in exploring and finding solutions.	Problem Statement Client Target Market Goal Outcome	Write Problem Statements Clearly Define Real World Problems With Obtainable Goals	Identify Problem Lecture & Discussion Project Identification Assignment 1 on 1 Teacher Conversation & Questioning
Brainstorming Stage	Vary Depending on complexity of project	1.3.a Effective Research Strategies Students use effective research strategies to find resources that support their learning needs, personal interests and creative pursuits.	Research Ideate List Ideas Sketches	Research Potential Solutions Visually Communicate Ideas Through Technical Sketches	Brainstorming Lecture & Discussion Project Research & Planning Sheet Technical Sketches 1 on 1 Teacher Conversation & Questioning

Design Stage	Vary Depending on complexity of project	1.6.c Communicate Complex Ideas Students use digital tools to visually communicate complex ideas to others.	Focus Feasibility Digital Design CAD	Determine Feasibility of Ideas Narrow # of Ideas and Focus Formally & Technically Document Final Ideas Using Digital Software	Project Design Lecture & Discussion Digital Design On Computer Software 1 on 1 Teacher Conversation & Questioning
Build Stage	Vary Depending on complexity of project	1.4.c Prototypes Students develop, test and refine prototypes as part of a cyclical design process.	Manufacturing Process Assemble Create Safety	Execute The Building of a Solution Based On Prior Planning Safety Build and Use Machines	Project Building & Safety Lecture & Discussion Cost Analysis WS Building Plan
Test & Evaluate Stage	Vary Depending on complexity of project	1.1c Feedback to Improve Practice Students use technology to seek feedback that informs and improves their practice and to demonstrate their learning in a variety of ways.	Analyze Evaluate	Analyze Results Evaluate if Solution Solves Problems	Project Evaluation Lecture & Discussion Data Collection Activity
Redesign Stage	Vary Depending on complexity of project	1.4.c Prototypes Students develop, test and refine prototypes as part of a cyclical design process.	Reverse Engineering	Determine Need for Any Changes in the Final Product Based on Observations and Data Collected During Test and Evaluation	Project Success Questionnaire Redesign Needs Assessment
Share Solutions Stage	Vary Depending on complexity of project	1.6.d Customize the Message Students publish or present content that customizes the message and medium for their intended audiences.	Document Explain Portfolio Final Product	Reflect The Success of the Project, and The Learning You Did During the Process of Designing, Building & Testing Your Solution	Portfolio Presentation of Portfolio Final Discussion

*Students will repeat the above progression a varying number of times depending on the number of projects they complete. That is determined by the size and complexity of problems they are solving or projects they work on.